

Calc 1 – Applications of the Integral

Worksheet A

1. Find the area between $y = x^2$ and $y = 2x$ on $[0, 2]$.
2. The region under $y = \sqrt{x}$, $x \in [0, 4]$, is revolved about the x -axis. Find the volume (disks).
3. Find the arc length of $y = \frac{2}{3}x^{3/2}$ on $[0, 3]$.
4. Find the average value of $f(x) = \sin x$ on $[0, \pi]$.
5. Solve the separable ODE $\frac{dy}{dx} = xy$, $y(0) = 2$.

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Worksheet B

1. Find the area enclosed between $y = x^2$ and $y = x + 2$.
2. The region under $y = x$, $x \in [0, 1]$, is revolved about the y -axis. Find the volume (shells).
3. Find the arc length of $y = \ln(\cos x)$ on $[0, \pi/4]$.
4. A force $F(x) = 10 - x$ acts on an object moving from $x = 0$ to $x = 5$. Find the work done.
5. Solve the IVP $\frac{dy}{dx} = \frac{y}{x}$, $y(1) = 3$ (with $x > 0$).

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Answer Key (Worksheets A & B)

Worksheet A.

1. Intersection: $x^2 = 2x \Rightarrow x = 0, 2$. On $[0, 2]$, $2x \geq x^2$. $A = \int_0^2 (2x - x^2) dx = [x^2 - x^3/3]_0^2 = 4 - 8/3 = \boxed{4/3}$.
2. $V = \pi \int_0^4 (\sqrt{x})^2 dx = \pi \int_0^4 x dx = \pi \cdot 8 = \boxed{8\pi}$.
3. $y' = \sqrt{x}$. $L = \int_0^3 \sqrt{1+x} dx = [\frac{2}{3}(1+x)^{3/2}]_0^3 = \frac{2}{3}(8-1) = \boxed{14/3}$.
4. $\bar{f} = \frac{1}{\pi} \int_0^\pi \sin x dx = \frac{1}{\pi} \cdot 2 = \boxed{2/\pi}$.
5. Separate: $\frac{dy}{y} = x dx \Rightarrow \ln|y| = \frac{x^2}{2} + C \Rightarrow y = Ae^{x^2/2}$. $y(0) = 2 \Rightarrow A = 2$. $\boxed{y = 2e^{x^2/2}}$.

Worksheet B.

1. Intersect: $x^2 = x+2 \Rightarrow x = -1, 2$. $A = \int_{-1}^2 (x+2-x^2) dx = [x^2/2+2x-x^3/3]_{-1}^2 = \frac{10}{3} - (-\frac{7}{6}) = \boxed{9/2}$.
2. Shells about y -axis: $V = 2\pi \int_0^1 x \cdot x dx = 2\pi \cdot \frac{1}{3} = \boxed{2\pi/3}$.
3. $y' = -\tan x$, $1 + (y')^2 = 1 + \tan^2 x = \sec^2 x$. $L = \int_0^{\pi/4} \sec x dx = [\ln|\sec x + \tan x|]_0^{\pi/4} = \boxed{\ln(\sqrt{2}+1)}$.
4. $W = \int_0^5 (10-x) dx = [10x - x^2/2]_0^5 = 50 - 12.5 = \boxed{37.5}$.
5. Separate: $\frac{dy}{y} = \frac{dx}{x} \Rightarrow \ln y = \ln x + C \Rightarrow y = Ax$. $y(1) = 3 \Rightarrow A = 3$. $\boxed{y = 3x}$.